FINMARK CORPORATION · DATA ANALYTICS

ML SOLUTION: EXPLORATORY DATA ANALYSIS & VISUALIZATION

Terminal Assessment — Preprocessing, Exploratory Analysis, and Data Visualization

Team 1

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MO-IT162 · Math For Machine Learning: Linear Algebra

INTRODUCTION

BACKGROUND & PURPOSE OF THIS ASSESSMENT

FinMark Corporation is preparing to apply machine learning to its growing base of customer data, but reliable models start with reliable data. This presentation is our Terminal Assessment for MO-IT162 — a single, mentor-reviewed account of the preprocessing, exploratory analysis, and visualization work our team completed across Milestone 1 and Milestone 2.

Preprocessing & EDA

Cleaning three linked datasets and surfacing early patterns in Milestone 1.

Data Visualization

Turning those patterns into distribution, sentiment, relationship, and trend charts in Milestone 2.

Verification & Revisions

Resolving flagged anomalies and sharpening our conclusions based on mentor feedback.

PROJECT OVERVIEW

WHY THIS MATTERS TO FINMARK

FinMark Corporation needs reliable customer data to support three upcoming machine learning initiatives. This Milestone reviews and visualizes the Milestone 1 EDA results across three datasets Demographics, Transactions, and Social Media Interactions to confirm data quality before model development begins.

Customer Segmentation

Grouping customers by age and behavior for targeted strategies.

Purchase Behavior Prediction

Forecasting spending patterns from transaction history.

Sentiment Analysis

Classifying customer opinions from social media interactions.

DATA & METHODOLOGY

THREE DATASETS, ONE CUSTOMER VIEW

Dataset	Key Fields	Role in Analysis
Demographics	CustomerID, Age, Gender, IncomeLevel	Customer profile and Age distribution
Transactions	CustomerID, Amount_Clean, ProductCategory, TransactionDate	Spending behavior and trends over time
Social Media Interactions	CustomerID, Platform, InteractionType, Sentiment	Customer sentiment and engagement

All three datasets share CustomerID, allowing Demographics and Transactions to be merged for relationship analysis.

Pandas
Load, clean, group, merge

NumPy
Descriptive statistics

Matplotlib
Bar & line charts

Seaborn
Histograms, scatter, heatmap

DISTRIBUTION & FREQUENCY ANALYSIS

TWO KEY VARIABLES, TWO DATA-QUALITY FLAGS

Customer Age

45.4	45	19.7
Mean	Median	Std. Dev.

63 anomalous records at -1 and 150 require investigation

Transaction Amount

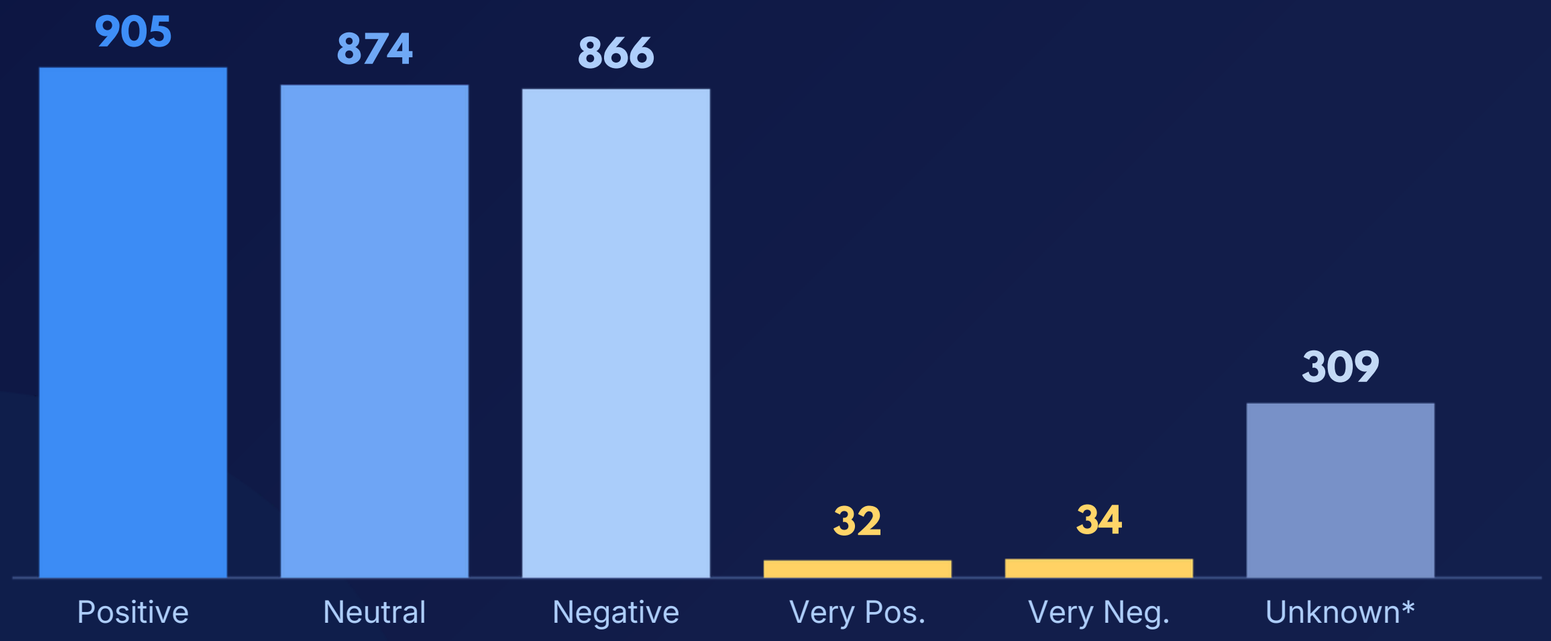
496.6	518.5	228–747
Median	IQR	25th–75th pct.

70 transactions ≤ 0, including a recurring -100

Both distributions are well-behaved at the center, but the extreme values at either end need to be resolved before modeling.

DISTRIBUTION & FREQUENCY ANALYSIS

SENTIMENT CLASSES ARE IMBALANCED



*Unknown (309 records) is imputed missing sentiment, not a genuine customer opinion — Very Positive and Very Negative are genuinely rare, risking weak model performance on those classes.

RELATIONSHIP & TREND ANALYSIS

AGE DOESN'T PREDICT SPENDING

0.021

Correlation: Age × Transaction Amount

Confirmed by both the scatter plot and the correlation heatmap essentially no linear relationship

Average Amount by Product Category

470	510
Health & Beauty (lowest)	Home & Garden (highest)

All five categories fall within this narrow range no single category dominates average spend.

FINMARK

RELATIONSHIP & TREND ANALYSIS

A LATE-2024 DECLINE WORTH INVESTIGATING



September and October 2024 recorded the two lowest monthly averages on record. This pattern is worth investigating further — it does not, on its own, prove a lasting shift in customer spending behavior, since seasonality, transaction mix, or external business factors could also explain it.

OUTLIERS & ANOMALIES

WHAT THE CHARTS REVEAL ABOUT DATA QUALITY

Age Anomalies

63 records at -1 or 150, likely data-entry errors or placeholders.

Resolved: masked & median-imputed — 0 remaining (mean 45.4→44.7, std 19.7→14.4)

Nonpositive Amounts

70 transactions ≤ 0, possibly refunds, voided sales, or errors.

Isolated for review — mean rises from 489.8 to 504.3 when excluded

Unknown Sentiment

309 records are imputed placeholders, not genuine opinions.

Flagged as its own category, not dropped, to avoid biasing results

The box plot did not flag the 70 nonpositive transactions as statistical outliers — proof that automated outlier detection alone cannot determine whether a value is valid. Domain knowledge must be applied alongside the statistics.

VISUALIZATION INSIGHTS

WHAT THIS MEANS FOR FINMARK'S ML ROADMAP

Segmentation

Age alone is a weak signal — segmentation models need a combination of variables.

Sentiment Classification

Class imbalance means a model may struggle on Very Positive and Very Negative without resampling or class weighting.

Data Readiness

Anomalies must be resolved and validated before training — unresolved issues risk biased predictions.

Overall, the visualizations confirm that FinMark's datasets are reliable at their core but require targeted cleanup before supporting customer segmentation, purchase behavior prediction, and sentiment classification.

RECOMMENDATIONS & NEXT STEPS

BEFORE FINMARK MOVES TO MODELING

1. Investigate and resolve the 63 Age and 70 Transaction Amount anomalies, with documented before-and-after validation.
2. Apply resampling or class weighting to address the Sentiment class imbalance before classification.
3. Investigate the September–October 2024 decline against seasonal and business factors before concluding a behavior shift.
4. Maintain a clear, documented verification trail so future milestones can trace every data-quality decision.

Visualization confirmed and sharpened our Milestone 1 findings. With these items addressed, FinMark's datasets will be ready to support reliable machine learning development.